MILITARY SPECIFICATION
INDICATOR, CONTROL C-10250A/ALR-67(V)

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The equipment covered by this specification is the C-10250/ALR-67(V) Control Indictor, the primary operator interface for mode selection and control to the AN/ALR-67(V) Countermeasures Receiving Set.

1.2 Associated equipment. The equipment operates with the associated equipment listed in 6.8 and as shown on Figure 1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation; Packaging, Methods of
MIL-W-5088	Wiring; Aerospace Vehicle

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Dept (SESD) Code 93, Lakehurst, N.J. 08733-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-E-6051	Electromagnetic Compatibility Requirements, System
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment
MIL-T-23103	Thermal Performance Evaluation, Airborne Electronic Equipment and Systems, General Requirement for
MIL-M-38510	Microcircuits, General Specification for
MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded and Breech Coupling), Environmental Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-C-85332	Computer, Countermeasures, CP-1293/ALR-67(V) and Azimuth Indicator IP-1276/ALR-67(V)
MIL-R-85566	Receiver, Radar (Special) R-2055/ALR-67(V)
MIL-R-85567	Antenna (Integrated Array) AS-3190/ALR-67(V)
MIL-R-85568	Receiver, Radar (Quadrant) R-2148/ALR-67(V)

Space and Naval Warfare Systems Command

WS-6536	Process Specification, Procedures and Requirements for Preparation and Soldering of Electrical Connections

Naval Air Systems Command

EI-656	Naval Air Systems Command Avionics Installation Instructions for Receiving Set Countermeasures AN/ALR-67(V)2 Configuration

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability/Verification/Demonstration/ Evaluation of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-965	Parts Control Program
MIL-STD-1562	List of Standard Microcircuits
DOD-STD-1686	Electrostatic Discharge Control Program
MIL-STD-2076	Unit Under Test Compatibility With Automatic Test Equipment, General Requirements for
MIL-STD-2077	Test Program Sets, General Requirements for
MIL-STD-2084	Maintainability of Avionic and Electronic Systems and Equipment, General Specification for
MIL-STD-45662	Calibration System Requirements

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HANDBOOKS

Military

DOD-HDBK-263 Electrostatic Discharge Control Handbook

(Copies of specifications, standards and handbooks, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

3. REQUIREMENTS

3.1 Quality inspection. The equipment procured under this specification shall be qualified to the extent specified in the contract and section 4.

3.2 Reliability. The contractor shall conduct a reliability program in accordance with MIL-STD-785 and the contract. (See 6.2.2)

3.2.1 Operational Stability. The equipment shall operate with specified performance, continuously or intermittently for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible during equipment operation.

3.2.2 Operating life. The equipment shall have a total operating life of 10,000 hours with minimal servicing and replacement of parts.

3.2.3 Reliability in mean time between failure (MTBF). The equipment shall have an MTBF equal to12,500 hours (theta one). Equipment meeting this specification shall have been qualified to 4.2.4 and screened in accordance with 4.3.4. The MTBF (theta one) value shall only be applicable for a discrimination ratio of 2. The MTBF shall not include indicator lamps located in the control panel pushbutton switches. Burn-out of these lamp bulbs shall not constitute equipment failure.

3.3 First article. When specified in the contract or purchase order a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.4 Parts and assemblies. In the selection of parts and assemblies, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall apply:

a. Microelectronic items shall conform to the requirements specified herein.

b. Electronic piece parts shall conform to the requirements for ER, JAN-TX or MIL-M-38510, Class B, as applicable to the part type. Electronic piece parts not conforming to the above shall be considered nonstandard, and shall be approved in accordance with 3.4.1. below. The order of precedence for parts other than electronic parts shall be as follows.

(1) Military standard parts.

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(2) Military specification parts (not listed in book form standards).

(3) Nonstandard parts previously approved for use on AN/ALR-67(V) and AN/ALR-45F(V).

(4) Nonstandard parts (require nonstandard parts approval).

c. Nonrepairable assemblies shall be used in accordance with MIL-STD-2084 and as specified in MIL-E-5400.

d. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

e. Non electronic items shall conform to the requirements of MIL-E-5400.

3.4.1 Nonstandard parts and materials approval. Approval for the use of nonstandard materials and parts shall be obtained as outlined in the contract. Specific design, process control, screening, quality conformance/qualification test requirements for design critical microelectronic hybrid devices shall be in accordance with the provisions delineated in each applicable procuring activity approved control drawing. Monolithic microelectronic devices shall be approved as specified in MIL-STD-965, except that approval or interchangeability demonstration shall not be required for devices which are: (See 6.2.2)

a. Listed in MIL-STD-1562.

b. Available from at least two MIL-M-38510 QPL-listed sources.

3.5 Design and construction. The equipment shall conform to all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein and in the contract. (See 6.2.1) Soldering requirements shall be in accordance with WS-6536. An electrostatic discharge (ESD) control program shall be incorporated in accordance with DOD-STD-1686 and DOD-HDBK-263.

3.5.1 Control panel. The equipment control panel shall conform to the applicable requirements of MIL-C-6781. The configuration of the panel shall be approved by the procuring activity prior to preproduction testing. (See 6.2.2)

3.5.2 Interchangeability. The equipment shall meet the interchangeability requirements of MIL-E-5400.

3.6 Provisions for maintainability. The contractor shall conduct a production maintainability program in accordance with MIL-STD-470.

3.6.1 Automatic test equipment (ATE). The equipment shall have WRA and SRA level test point provisions in accordance with MIL-STD-2084 to provide test point compatibility and usage with ATE to preserve the Intermediate level requirements of 3.6.3.1 and 3.6.3.2 specified herein.

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a. Requirement 101 — Maintainability Program - In Paragraph 4.3 (Maintainability prediction), the reference to Procedure 1 of MIL-HDBK-472 shall be changed to Procedure 2 of MIL-HDBK-472.

b. Requirement 102 — Failure Mode and Effects Analysis - In Paragraph 4 (FMEA, the requirements of MIL-STD-1629 shall be considered analysis at the "functional circuit" level and not at the "piece part" level.

c. Requirement 103 — Physical Design - As follows:

(1) In Paragraph 4 (Design), the requirement for Level of Repair (LOR) analysis shall be deleted.

d. Requirement 104 — Built-in-Test - As follows:

(1) In Paragraph 5.2.2 (BIT fault isolation level), the third sentence shall be deleted in its entirety.

(2) Table 104-I (Non-ambiguous SRA isolation groups) shall be deleted in its entirety.

(3) BIT calibration if applicable, shall reside within the CP-1293 when connected as a system.

(4) In Paragraph 11.2 (fault indicators), the third and fourth sentences shall be deleted.

3.6.2 Organizational level maintenance requirements.

3.6.2.1 Interchangeability. Replacement of equipment at the Organizational level shall be possible without adjustment or alignment of internal controls.

3.6.2.2 Built-in test (BIT). The following performance requirements for BIT shall be in accordance with MIL-STD-2084 and shall apply when the C-10250/ALR-67 is operated with the remainder of the ALR-67(V) as shown on Figure 1.

3.6.2.2.1 Corrective action time. The characteristics of the equipment shall be such that it will be possible in 95 percent of the cases of failure to perform all corrective Organizational level maintenance actions, other than combat damage repair, within a period not exceeding 30 minutes (the 30 minutes corrective action time shall include all elements, except aircraft access time). As here applied, Organizational level maintenance action shall include the following:

a. Verification of a fault.
b. Location of the fault to a Weapon Replaceable Assembly (WRA).
c. Correction or repair.

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d. Adjustment or alignment (when required).
e. Checkout of the repair.

3.6.3 Intermediate level maintenance requirements. Maintenance activity at the Intermediate level shall consist of: (1) Repair of the equipment by fault isolation to the faulty SRA; (2) removal and replacement of the faulty SRA; and 3) verification of repair completion by complete end-to-end test. The design shall permit repair of the faulty SRA at the Intermediate level repair facilities; i.e., module or microelectronics shop. The equipment shall be composed of Shop Replaceable Assemblies (SRAs). SRA's are defined as discrete replaceable modules such as circuit cards, traveling wave tube assemblies or modular (other than discrete) components. The term SRA includes quick replaceable assemblies (QRAs) and bench replaceable assemblies (BRAs) as defined in MIL-STD-2084. Not less than 67 percent of all SRAs within the equipment shall be of the quick replaceable assembly (QRA) type as specified in MIL-STD-2084.

3.6.3.1 Intermediate level fault isolation capability. The ratio of the number of SRAs directly isolated without ambiguity by the AN/USM-458(V) Test Set and the ALR-67(V) Operational Test Procedures (OTPs) to the total number of SRAs shall be 1.0 as a design goal. The minimum acceptable requirements for non-ambiguous SRA isolation shall be as follows:

a. In at least 90 percent of the cases of probable malfunction of an SRA, the fault shall be isolated to that SRA.

b. In 95 percent or more of the cases of probable malfunction of an SRA, the fault shall be isolated to that SRA and no more than one other SRA.

c. In all cases of probable malfunction of an SRA, the fault shall be isolated to that SRA and no more than two other SRAs.

3.6.3.1.1 Fault isolation to the part level. Each SRA of the quick replaceable assembly or bench replaceable assembly type shall contain shop repair test points to quantitatively determine the performance level of the SRA and, in the event of failure, to diagnose which nonrepairable or small group of nonrepairables within the SRA is at fault. The shop repair test points shall provide the means for:

a. Measuring the necessary input/output and circuit parameters to enable isolation of the faulty nonrepairable or small group of nonrepairables as follows:

(1) When the SRA contains 10 or less nonrepairables, isolation to groups of two or less shall be possible for 50 percent of possible faults. Isolation to groups of four or less must be possible for all possible faults.

(2) When the SRA contains more than 10 nonrepairables, isolation to groups of four or less shall be possible for 80 percent or more of the possible faults. Isolation to groups of ten or less must be possible for 100 percent of the possible faults.

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b. Measurement of necessary input/output parameters to enable calibration or alignment of the SRA.

3.6.3.2 Intermediate level mean time to repair (MTTR). The mean time to repair a failed WRA at the Intermediate level shall not exceed 45 minutes. The MTTR shall include fault isolation to the SRA, removal and replacement of the faulty SRA, and verification of WRA repair by means of an end-to-end test to ensure the return to inventory condition of the WRA. The maximum time to repair, within which 95 percent of all Intermediate level repair actions are required to be accomplished, shall not exceed 2.5 hours. Fault isolation of the failed SRA may be accomplished with removal of the top cover from the equipment. Fault isolation shall include the time to remove covers from the WRA, isolate the faulty SRA, remove and replace the faulty SRA, replace WRA covers and complete end-to-end test to ensure the return to inventory condition of the reworked WRA.

3.6.3.2.1 Removal and replacement provisions. The design of the equipment shall facilitate removal and replacement of the SRAs listed below.

a. PROM (Dummy) Assembly (A1)
b. PROM Control Assembly (A2)
c. Chassis

3.6.3.3 Interchangeability. SRA interchangeability shall be in accordance with the shop fixed-interface ratio specified in MIL-STD-2084. The ratio of the number of SRAs which do not require adjustment or trimming after installation within the WRA to the total number of SRAs within the WRA shall be 1.0.

3.6.3.4 Depot level maintenance requirements. The Depot level repair activity shall consist of all actions beyond the capability of the Intermediate level required to restore the equipment to acceptable operating condition.

3.6.3.5 Encapsulation. Repairable SRAs, assemblies and modules shall not be encapsulated, potted or embedded.

3.6.3.6 Fastners. The different types of fasteners (screws, bolts, quick-release fasteners) used in the equipment shall be minimized to reduce tool requirements for maintenance.

3.6.3.7 Maintenance demonstration. Maintainability verification and demonstration shall be required in accordance with 4.2.6 herein.

3.7 Safety. The design of the C-10250/ALR-67(V) Indicator, Control shall incorporate the safety requirement of MIL-STD-454, Requirement 1. (See 6.2.2)

3.8 Performance characteristics.

3.8.1 Performance. Values set forth to establish the requirements of specified performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions shall be specified. The performance of the equipment shall not be degraded, by operational interface with the CP-1293/ALR-67(V) Countermeasures Computer or the R-2055/ALR-67(V) Radar Receiver (Special).

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3.8.2 Operating power. The equipment shall meet all applicable requirements of MIL-STD-704, Category B and shall give specified performance when energized from the following power sources having characteristics and limits as specified in MIL-STD-704. The power required shall not be greater than the specified amounts.

a. 28 +3 VDC, 30 watts maximum
b. 12 ±0.6 VDC, 3 ±1.2 watts

3.8.2.1 Lighting power. Input power for lighting the indicator lamps shall require no more than 1.0 ampere from 28-VDC aircraft power.

3.8.2.2 Degraded performance. The equipment shall operate at reduced performance within 18 to 22 VDC and operation is not required at less than 18 VDC or greater than 31 VDC . Degraded performance shall be permitted for all voltage transients from 0 to 0.08 seconds duration as shown in Figures 6 and 10 of MIL-STD-704, and performance shall not be affected by short pulses of less than 50 microseconds duration. Degraded performance for undervoltage conditions from 0 to 7 seconds duration, shall be permitted as shown in Figures 7 and 12 of MIL-STD-704. Equipment operation shall return to normal without requiring the equipment to time-out when these voltage transients expire. Program data and register contents shall not be altered by degraded performance. When power returns to normal, controls required to commence orderly processing at the program starting location shall be automatically reset. Operations shall return to normal with no resulting damage to the equipment.

3.8.2.3 Overload protection. Overload protection shall be provided external to the equipment.

3.8.2.4 Undervoltage protection. The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.8.2.5 Mechanical interlocks. There shall be no mechanical interlocks included as part of the equipment.

3.8.3 Cooling. The design for cooling the equipment shall comply with the requirements of MIL-E-5400. No cooling air shall be provided for the equipment.

3.8.4 Operation. This WRA is the primary operator control interface for the AN/ALR-67(V) Countermeasures Receiving Set. The equipment shall contain a total of five pushbutton switches, one rotary switch and two control potentiometer knobs as shown on Figure 2. The pushbutton switches shall be back-lighted, alternate-action controls. When the power switch is depressed, the back-lighted on legend shall be illuminated and the physical position of the power switch shall indicate that the equipment is operating. The power switch upper surface shall protrude 1/8 inch during power off conditions and shall be flush during power on conditions. The following switch-indicator legends also shall be illuminated when the power switch is enabled (see Figure 2):

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a. DISPLAY (lower legend)
b. SPECIAL (lower legend)
c. OFFSET (lower legend)
d. BIT (lower legend)

The upper legends of the split-legend special, offset and bit switches shall be illuminated via lamp driver inputs L1, L2 and L7. The upper legend of the split legend display switch shall be illuminated by self-latching circuitry within the switch. Pushbutton switch operation and output to the serial data bus is specified in greater detail in 3.8.7 and Table I. The display type rotary switch shall select one of five available positions (F, U, A, I and N) which shall cause data to be placed on the serial data bus, as specified in 3.8.5 and 3.8.7.6, when the serial data bus is queried. The audio control knob shall be used to vary the amplitude of audio input signals between 20 and 80 percent. The dimmer control knob shall be provided to vary the legend lamp intensity from dim to bright. A dummy Programmable Read Only Memory (PROM) circuit card assembly shall be provided with the equipment. This card is furnished for future emitter library expansion and, shall not contain any traces or electrical devices. The dummy PROM circuit card assembly shall be easily removable from the faceplate of the equipment. Overall equipment functional operation is shown on Figure 3 and in 3.8.5, 3.8.6 and 3.8.7.

3.8.5 Equipment functional description. The C-10250/ALR-67(V) Control Indicator, when connected in a system configuration as an integral part of the AN/ALR-67(V) Countermeasures Receiving Set, shall provide interface to the CP-1293/ALR-67(V) Countermeasures Computer and the R-2055/ALR-67(V) Radar Receiver (Special) over a shared serial data bus. The equipment shall also provide interface with other discrete signals. Equipment operation in the AN/ALR-67(V) system shall enable the operator to energize the CP-1293 Computer and the R-2055 Special Receiver by depressing the power pushbutton switch. Indication of power being applied to these two units shall be evident by illumination of the pushbutton switch on legend. Operation of all other equipment controls and indicators shall be dependent upon actuation of the power switch.

3.8.5.1 Display switch. The display pushbutton switch shall be utilized by the CP-1293/ALR-67 Computer to limit the number of threat symbols displayed by the IP-1276/ALR-67(V) Azimuth Indicator to the six highest priority threats. When the switch is depressed, the computer shall command the azimuth indicator to display the "L" symbol (for limit) in the display status area, informing the operator that the limit feature is enabled. Also, when the switch is depressed, the light bulbs in the upper limit legend shall be illuminated by self-latching circuitry within the switch, indicating to the operator that the limit feature is enabled. Depressing the display switch when the limit option is active shall return the system to normal operations allowing up to 15 threats to be displayed, shall extinguish the upper limit legend, and shall remove the "L" symbol from the azimuth indicator display status area.

3.8.5.2 Special switch. The special pushbutton switch shall be used by the CP-1293/ALR-67 Computer to provide the operator with an expanded view (through the IP-1276 Azimuth Indicator) of built-in test status in the AN/ALR-67(V) system during manual BIT. When the switch is depressed and manual BIT is activated, the upper enable legend shall be illuminated to indicate fail-code readout displayed by a 3-by-4 character matrix on the azimuth indicator screen.

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3.8.5.3 Offset switch. The offset pushbutton switch shall permit the operator to enable or disable the offset feature in the IP-1276/ALR-67 Azimuth Indicator. When the CP-1293/ALR-67 Computer senses the offset switch being depressed, and if the offset feature is disabled, the computer shall enable the offset feature. Offset enable shall cause automatic threat symbol separation in the IP-1276/ALR-67 Azimuth Indicator, which shall spread overlapping threat emitter symbols that are difficult to distinguish from each other on the azimuth indicator display screen. Simultaneously with threat symbol separation, the computer shall command the equipment to illuminate the upper enable legend on the pushbutton switch, informing the operator that the offset feature is enabled. Conversely, if the offset feature is enabled and the offset switch is depressed, the computer shall command the upper enable legend to extinguish and shall command the azimuth indicator to return all threat symbols to their measured bearing.

3.8.5.4 BIT switch. The BIT pushbutton switch shall be utilized to initiate manual BIT in the AN/ALR-67(V) system. When a failure is detected by continuous BIT, the upper fail legend on the switch shall be illuminated (under control of the CP-1293/ALR-67 Computer). When the switch is depressed and manual BIT is activated, the light bulbs in the BIT switch upper legend (fail) shall be illuminated in a blinking manner (under control of the CP-1293/ALR-67 Computer), indicating to the operator that BIT is in process. The fail lamp shall be extinguished when BIT failures are cleared.

3.8.5.5 Display type switch. The display type rotary switch shall be utilized to select one of five threat classes for display on the IP-1276/ALR-67(V) Azimuth Indicator as specified in 3.8.7.6.

3.8.5.6 Audio control. The audio knob shall be utilized by the operator to vary the volume of audio alert status tones in the AN/ALR-67(V) system. When the knob is turned in the clockwise direction, the volume shall increase. When the knob is turned in the counterclockwise direction, the volume level shall decrease.

3.8.5.7 Dimmer (DMR) control. The dimmer knob shall be utilized by the operator to vary the illumination intensity of light bulbs back-lighting the five pushbutton switches to provide clear legend visibility in daylight and night-light conditions. When the knob is turned in the clockwise direction, the light bulb illumination intensity shall increase. When the knob is turned in the counterclockwise direction, the illumination intensity shall decrease.

3.8.6 Control indicator C-10250/ALR-67(V). The equipment shall be identical for the AN/ALR-67(V)1 and AN/ALR-67(V)2 system configurations shown on Figure 1, and shall meet the requirement specified in 3.8.7.

3.8.7 Controls and indicators. The equipment control switches and indicator lamps shall conform to the requirements of Figure 3. The five pushbutton switches shall be alternate-action, plug-in units which may be directly replaceable with the equipment installed in the aircraft. The power switch shall be mechanically latching as shown on Figure 3. All other pushbutton switches shall be mechanically momentary and nonlatching. Each unit shall be replaceable without removing the other four pushbutton switches or the equipment faceplate. The light bulbs shall be easily removable from the rear of each switch unit once the switch unit has been removed from the equipment.

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3.8.7.1 Power switch. The power pushbutton switch, when depressed, shall close the signal path from Connector J1, Pin 9 to 10 and Pin 4 to 34 (providing switched operating power to the CP-1293/ALR-67 Computer and R-2055 Special Receiver, respectively, when the equipment is connected in a system configuration). The back-lighted on legend of the power switch shall illuminate when the path from Pin 4 to 34 is closed. The lower legends of the split-legend display, special, offset and BIT switches shall also illuminate when the power switch is enabled

3.8.7.2 Display switch. The display pushbutton switch, when depressed as the serial data bus is queried, shall cause an active indication in the twelveth position of the serial data bus (see Table I). The upper limit legend of the display switch shall illuminate whenever the switch is active (enabled by the operator).

3.8.7.3 Special switch. The special pushbutton switch, when depressed as the serial data bus is queried, shall cause an active indication in the thirteenth position of the serial data bus (see Table I). The upper enable legend of the special switch shall illuminate when an active signal is placed on Connector J1, Pin 5. The lower legend shall illuminate as specified in 3.8.7.1.

3.8.7.4 Offset switch. The offset pushbutton switch, when depressed as the serial data bus is queried, shall cause an active indication in the fourteenth position of the serial data bus (see Table I). The upper enable legend of the offset switch shall illuminate when an active signal is placed on Connector J1, Pin 22. The lower legend shall illuminate as specified in 3.8.7.1.

3.8.7.5 Built-in-test switch. The BIT pushbutton switch, when depressed as the serial data bus is queried, shall cause an active indication in the eleventh position of the serial data bus (see Table I). The upper fail legend of the BIT switch shall illuminate when an active signal is placed on Connector J1, Pin 6. The lower legend shall illuminate as specified in 3.8.7.1.

3.8.7.6 Display type switch. The dis type rotary switch shall cause an active indication as follows (depending on its position) upon querying of the serial data bus:

a. F Position: When the display type switch is in the F position, an active indication shall be caused in the fourth position of the serial data bus (see Table I).

b. U Position: When the display type switch is in the U position, an active indication shall be caused in the fifth position of the serial data bus (see Table I).

c. A Position: When the display type switch is in the A position, an active indication shall be caused in the sixth position of the serial data bus (see Table I).

d. I Position: When the display type switch is in the I position, an active indication shall be caused in the seventh position of the serial data bus (see Table I).

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e. N Position: When the display type switch is in the N position, an active indication shall be caused in the eighth position of the serial data bus (see Table I).

3.8.7.7 Audio control. The audio control knob shall provide amplitude control of a signal (not greater than 10 volts peak and less than 2 KHz in frequency) input at Connector J1, Pin 15 and output at Pin 14. The input signal shall be capable of driving a load of not less then 5 kilo ohm. The output at Connector J1, Pin 14 shall be capable of driving a 600-ohm load, and shall decrease when the audio knob is turned in the counterclockwise direction.

3.8.7.8 Dimmer control. The dimmer control knob shall simultaneously vary the illumination intensity of all pushbutton switch light bulbs. The light bulb intensity shall increase when the dimmer knob is turned in the clockwise direction, and it shall decrease when turned in the counterclockwise direction.

Table I Serial Data Bus Switch/BIT Postion

SWITCH (position)	ACTIVE BIT INDICATION POSITION															
	1*	2*	3*	4	5	6	7	8	9*	10*	11	12	13	14	15*	16*
display type Switch (turned to)																
F (Friendly)				X												
U (Unknown)					X											
A (Anti-aircraft)						X										
I (Intercept)							X									
N (Normal)								X								
BIT Switch (depressed)											X					
DISPLAY Switch (depressed/latched)												X				
SPECIAL Switch (depressed)													X			
OFFSET Switch (depressed)														X		

*Positions 1, 2, 3, 9, 10, 15 and 16 not used.

3.8.8 Interface requirements. The equipment interface to the AN/ALR-67(V) Countermeasures Receiving Set shall be as shown on Figures 1 and 2. Equipment interface to individual WRAs within the system shall be as described below.

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a. MIL-C-85332(AS) CP-1293/ALR-67(V) Countermeasures Computer (SECRET)
b. MIL-R-85566(AS) R-2055/ALR-67(V) Radar Receiver (Special) (SECRET)

3.8.8.1 Equipment/computer interfaces. The equipment interface with the CP-1293/ALR-67(V) Computer shall be as shown on Figure 4.

3.8.8.2 Equipment/special receiver interface. The equipment interface with the R-2055/ALR-67(V) Radar Receiver (Special) shall be as shown on Figures 1, 3 and Figure 4.

3.8.9 Mounting. Aircraft installation and mounting shall be in accordance with the applicable Electronics Installation Instruction EI-656.

3.8.10 Form factor. The equipment shall be enclosed as shown on Figure 5.

3.8.11 Contents. The equipment shall consist of six control switches and two control potention meters enclosed in a chassis with top an bottom covers. Major equipment components (classified as SRAs) shall be as listed in 3.6.3.2.1.

3.8.12 Electrical connections. The equipment connections to external circuits of the CP-1293/ALR-67(V) Computer (and to R-2055/ALR-67(V) Radar Receiver (Special) where noted) shall be provided via a crimp-type electrical connector plug (MS27467T15F35P) as specified in Table II.

Table II. (U) Connector J1 Pin Assignment

Pin No.	Function
4	+28 Vdc (ACFT Pwr)
5	L1 Indicator Driver (SPECIAL)
6	L7 Indicator Driver (FAIL)
9	Filtered 115 Vac, 400 Hz
10	On/Off Switches 115 Vac, 400 Hz
12	+5 Vdc Test
13	+12 Vdc
14	Level Controlled Audio (to ACFT Intercom)
15	Audio Tones
16	Serial Data Bus Enable Return
17	Serial Data Bus Enable
18	Serial Data Bus (R-2055/CP-1293)
21	+28 Vdc Return (ACFT Pwr)
22	L2 Indicator Driver (OFFSET)
24	Filtered 115 Vac, 400 Hz Return
25	On/Off Switched 115 Vac, 400 Hz Return
27	+12 Vdc Return
28	GND
30	Serial Data Bus Return (R-2055/CP-1293)
33	Switched +28 Vdc Return (R-2055)
34	Switched +28 Vdc (R-2055)

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3.8.13 Cabling and connections.

3.8.13.1 Cables and connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400, except that connectors shall be in accordance with MIL-C-38999, Series I. Connectors shall not be cadmium-plated to prevent dissimilar metals corrosion with aircraft connectors.

3.8.13.2 Interconnection cabling. The equipment shall be capable of specified operation using external wiring in accordance with the applicable requirements of MIL-W-5088. External wire bundles in the equipment external cable shall be shielded to meet interference control requirements. An external cable (W1) and 37-pin connector plug (J1), enabling input/output for power and operational control, shall be supplied as part of the equipment.

3.8.14 Warm-up time. The maximum warm-up time for the equipment shall be not greater than 2 minutes at standard conditions, and 5 minutes at service conditions.

3.9 Human factors. The human factors requirements for the design of the C-10250/ALR-67(V) Indicator Control shall be in accordance with MIL-E-5400.

3.10 Environmental requirements.

3.10.1 Standard conditions. The following conditions shall be used to establish performance characteristics under standard conditions and for making laboratory bench tests:

a.	Temperature:	+18° to +35°C
b.	Altitude:	Ground elevation of test facility
c.	Vibration:	Normal ambient conditions existing at test facility
d.	Humidity:	Ambient (Not to be greater than 90 percent relative)
e.	Input Voltage:	Within the limits of MIL-STD-704, Category B power

3.10.2 Service conditions. The equipment shall be designed to operate as specified under any of the environmental or service conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.10.2.1 Vibration. The equipment shall operate as specified when subjected to the random vibration of Figure 6 for 4.5 hours along each of the three principal axes.

3.10.2.2 Temperature. The equipment shall operate as specified when exposed to the temperature conditions specified in MIL-E-5400, Table I for Class 2 equipment. The ambient temperature within the specified ranges may remain constant for long periods or may vary at a rate as high as 1.0 degree per second.

3.10.2.3 Altitude. The equipment shall meet the altitude conditions specified in Table I of MIL-E-5400 for Class 2 equipment, both for continuous operation

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and nonoperating environments. The altitude may remain constant for long periods or may vary at a rate as high as 0.5-inch of mercury per second.

3.10.2.4 Shock. The equipment shall withstand dynamic shock stress during handling, transportation and service use as specified in MIL-E-5400.

3.10.2.5 Humidity. The equipment shall withstand humidity levels during handling, transportation and service use as specified in MIL-E-5400.

3.10.2.6 Salt fog atmosphere. The equipment shall withstand exposure to salt-sea atmosphere as specified in MIL-E-5400.

3.10.2.7 Explosive atmosphere. The equipment shall operate safely in an explosive atmosphere as specified in MIL-E-5400.

3.10.2.8 Interference control. The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461: CE01, CE02, CE03, CE04, CS01, CS02, CS06, RE01, RE02, RS02 and RS03. (Requirements tailored by Table III).

Table III (U) RFI Susceptibility

Frequency (MHz)	Field Strength (V/m) (average)	Pulse Width (usec)	PRF (Hertz)
0.014 - 100	20	CW	CW
100 - 1000	200	0.2 - 15	100 - 7200
1000 - 3590	200	0.2 - 15	100 - 10000
3590 - 3700	216	0.2 - 2	100 - 1200
3700 - 8500	200	0.2 - 10	100 - 1200
8500 - 9600	242	0.2 - 2	100 - 12000
9600 - 12000	200	0.2 - 2	100 - 12000
12000 - 16000	200	0.2 - 2	100 - 2000
16000 - 17000	226	0.2 - 2	0.1 - 2000
17000 - 33000	200	0.2 - 2	0.1 - 2000
33000 - 34000	200	0.2 - 2	0.1 - 2000
34000 - 40000	200	0.2 - 2	0.1 - 2000

(Terminate low band antenna with 50 ohm load prior to test)

3.11 Total weight. The total weight of the equipment, including the dummy PROM circuit card and cable, shall not be greater than 2.0 pounds.

3.12 Color and finish. The color and finish of the C-10250/ALR-67 Control Indicator shall be in accordance with the general provisions of MIl-E-5400.

3.13 Nomenclature, nameplates and identification marking. Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307. (See 6.2.2)

3.13.1 Assembly, subassembly and component identification. The equipment item marking and method of identification at the assembly, subassembly and component level shall be in accordance with MIL-STD-130.

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3.14 Workmanship. The standard of workmanship shall be in accordance with MIL-STD-454, Requirement 9. Equipment quality characteristics shall be in accordance with the approved design documentation and supplemental manufacturing documentation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5 of this specification. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General. The contractor shall develop and carry out a Quality Program for testing, manufacturing, and maintaining the quality and performance of the configuration item(s) (CI) and components (to levels specified in Section 3 and 5 of this specification).

4.1.3 Surveillance of operations. All operations will be closely observed/ monitored to determine that practices, methods and procedures are being properly implemented and adhered to.

4.1.3.1 Government product inspection. Government inspections (including in process inspection) will be performed to ensure the quality of the items offered for acceptance. Failure of the contractor to promptly correct deficiencies (either discovered by him or of which he is notified) will be cause for suspension of acceptance. until corrective action has been made or until conformance of the items to specified criteria has been demonstrated. Suspension of acceptance shall be in effect.

4.1.4 Classification of inspections. The inspection requirements specified herein are classified as follows.

a. First Article Tests (4.2)
b. Individual Acceptance (Screening) Tests (4.3)
c. Lot Sample Tests (4.4)

4.1.5 Test procedures/report. Test procedures shall include inspection requirements to assure quality control of the equipment. MIL-T-18303, MIL-T-23103 and MIL-STD-781 shall be used as applicable for preparation of test procedures for

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hardware. When approved test procedures are available, they will be furnished to the contractor and may be used on conditions authorized by the procuring activity. The procuring activity reserves the right to modify (increase, decrease or change) the test procedures to cost-effectively ensure that delivered CIs meet the requirements of this specification and the contract. (See 6.2.2)

4.1.5.1 Standard conditions. Unless otherwise specified herein, all examinations, inspections and measurements shall be performed in accordance with the following standard conditions and provisions:

a. Temperature: +18° to 35°C

b. Altitude: Ground elevation of test facility

c. Vibration: Normal ambient conditions existing at test facility

d. Humidity: Ambient (not to exceed 90 percent relative)

4.1.5.2 Test equipment and facilities. Test equipment and facilities shall be provided by the contractor, and shall be of sufficient accuracy, quality and quantity to permit performance of the examinations, tests and measurements specified herein. The limit of error in measurement apparatus shall not exceed 20 percent of the tolerance of the specified variable to be measured.

4.1.5.3 Calibration. All test and measurement equipment shall be calibrated and maintained in accordance with the requirements of MIL-STD-45662.

4.1.6 Presubmission testing. No CIs nor materials or parts shall be submitted to the Government inspector for final acceptance unless they have been subjected to test and inspection deemed necessary by the contractor and found to comply with the specification and the contract. When deviations from requirements are authorized by the procuring activity, the contractor shall keep a record of all CIs incorporating the deviations and notify the Government inspector when submitting the CIs for acceptance. When feasible, at the discretion of the Government inspector or procuring activity, presubmission testing and inspection may be participated in or witnessed by Government representatives to forestall unnecessary repeat testing and inspection when the CI, including software, is formally submitted for approval or acceptance.

4.2 First article inspection. First article tests shall be conducted by the contractor when ordered by the contract. (See 6.2.2)

4.2.1 Scope of tests. First article tests shall include all tests necessary to determine that the CIs meet all the requirements of this specification, other applicable specifications and the contract. In lieu of performing operational stability and life cycle testing (see 3.2.1 and 3.2.2), the contractor may use design data, the results of analysis or field experience with the AN/ALR-67. First article tests shall consist of the following as evidence of compliance: (See 6.2.2)

a. Physical Configuration Audit (see 4.2.2)
b. EMI/EMC Qualification Tests (see 4.2.3)
c. Environmental Qualification Tests (see 4.2.4)

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d. Reliability Growth Tests (see 4.2.5)
e. Maintainability Demonstration (see 4.2.6)

4.2.2 Physical configuration audit. Configuration inspection and interface testing specified below shall be performed on two of each equipments that have passed screening tests. (See 4.3)

a. Physical configuration audit
b. Physical interchangeability demonstration at the SRA level

4.2.3 EMI/EMC qualification tests. Electromagnetic interference/electromagnetic compatibility qualification tests shall be as specified in 4.2.3.1 and 4.2.3.2.

4.2.3.1 EMI. One of each equipment used in 4.2.3 tests shall be subjected to an EMI qualification test in accordance with MIL-STD-462, Class A1 equipment test requirements.

4.2.3.2 EMC. One complete AN/ALR-67(V) Countermeasures Receiving Set, incorporating the WRA described in this specification, shall be compatible for installation in fleet-configured aircraft for performance of EMC qualification test in accordance with MIL-E-6051.

4.2.4 Environmental qualification tests. One of each equipment used in tests shall be subjected to environmental qualification tests in accordance with Table IV. (See 6.2.2)

Table IV. Environmental Qualification Tests

Inspection or Test	Method/Procedure
Visual and Mechanical Inspection	4.3.2 herein
Production Acceptance Test	4.3.2 herein
Temperature/Altitude	MIL-STD-810, Method 520.2, Category 6 (Operating)
Shock	MIL-STD-810, Method 516.3, Procedure I (Non-operating)
Vibration	3.10.2.1 herein
Humidity	MIL-STD-810, Method 507.1, Procedure I (Non-operating)
Explosive Atmosphere	MIL-STD-810, Method 511.2 Operating)
Salt-Fog Atmosphere Procedure I	MIL-STD-810, Method 509.2, (Operating and Non-operating)

NOTE: The WRA shall be tested in accordance with approved test plans and procedures.

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4.2.5 Reliability Growth Tests (TAAF). Reliability Growth testing shall be performed in accordance with the contracts, and in accordance with approved test procedures with environmental cycles as shown on Figure 7. (See 6.2.2)

4.2.6 Maintainability demonstration test. Maintainability verification and demonstration test for I-level requirements shall be conducted in accordance with MIL-STD-2084 and Test Method (9) of MIL-STD-471. The maintainability of the diagnostic programs shall include verification of the accuracy and completeness of the functional (end-to-end) diagnostic test programs. This shall include insertion of faults into the equipment of the type and magnitude specified by MIL-STD-2077. The ability of the functional (end-to-end) diagnostic program to detect and identify the failure shall be verified prior to test. When specified in the contract or order, test requirement documents shall be prepared. (See 6.2.2)

4.3 Individual acceptance (screening) tests. The contractor shall be responsible for accomplishing all acceptance tests. Although all inspection and testing is subject to monitoring and witnessing by Government representatives, acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of acceptance of the finished CIs. (See 6.2.2)

4.3.1 Scope of tests. Acceptance tests shall include all tests specified in Table V to assure that production CIs are equal to the approved First Article in material, workmanship, performance and reliability and to correct deficiencies and incorporate changes. Acceptance tests shall include:

a. Production Acceptance Tests
b. Manufacturing screening

4.3.2 Production acceptance test. Production acceptance test of the requirements specified in Table V, including visual and mechanical inspection, shall be conducted in accordance with the approved acceptance test procedures. Production acceptance test shall be performed under standard ambient conditions delineated in 4.1.5.1 unless otherwise specified in the contract.

4.3.3 Manufacturing screening.

4.3.3.1 Random vibration. Each equipment shall be subjected to a minimum of ten minutes of 6 grms random vibration in accordance with the profile of Figure 8. The axis of excitation shall be that which is normal to the plane of the majority of printed circuit boards. Additional vibration may be used with approval of the procuring activity.

4.3.3.2 Thermal cycling. Each equipment shall be subjected to not less than 6 four hour thermal cycles ranging between -54 and +55°C. The temper- ature rate of change shall be not less than 5°C/minute. Additional cycles or more extreme temperatures may be used with the approval of the procuring activity. Random vibration may be combined with thermal cycling. The last 24 operating hours of thermal cycling shall be failure free.

4.4 Lot sample tests. Lot sample testing shall consist of quality conformance environmental tests (QCET) performed on equipment that has passed screening tests. The lot sample size and acceptance criteria shall be as specified in the contract (see 6.2.1). The quality conformance testing shall

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consist of temperature altitude in accordance with MIL-STD-810, Method 520.2 Cat 6 (operating), vibration in accordance with 3.10.2.1 herein, the equipment non operating.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in Table V.

4.6 Noncompliance. If a lot sample fails to pass inspection the contractor shall notify the procuring activity of such failure. The contractor shall take corrective action on the materials or processes, or both, as warranted, and on all unit of production which can be corrected, which are manufactured under essentially the same materials and processes and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the procuring activity has been taken. After the corrective action has been taken, inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the procuring activity). Final acceptance and shipment shall be withheld until reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the procuring activity.

5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or purchase order in accordance with MIL-P-116 and MIL-STD-129.

6. NOTES

6.1 Intended use. The equipment is intended for use in Naval aircraft to carry out specific military missions.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title number and date of this specification.
b. Reliability Program (see 3.2)
c. Design and construction, if other than specified in 3.5.
d. Levels of preservation-packaging and packing (see 5.1).
e. Lot sample sizes and acceptance criteria, if required.

6.2.2 (U) Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) incorporated into the Contract. When the provisions of DOD FAR Suppliment, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below

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Table V. (U) Quality Conformance Inspection

Requirement Description	Requirement Paragraph	First Article Tests	Production Acceptance Tests	Lot Sample Test	Test Paragraph
Reliability	3.2	X		X	4.2.6, 4.3.5
Manufacturing screening	3.2	X	X	X	4.2, 4.3.5 & 4.4
Interchangeability	3.5.2	X			4.2.3, 4.3.3
Maintainability Demonstration	3.6.3.7	X			4.2.7
Operating Power	3.8.2	X			4.2.5
Lighting Power	3.8.2.1	X			4.2.5
Degraded Performance	3.8.2.2	X			4.2.5
Overload Protection	3.8.2.3	X			4.2.5
Undervoltage Protection	3.8.2.4	X			4.2.5
POWER Switch	3.8.7.1	X	X	X	4.2, 4.3.3 & 4.4
DISPLAY Switch	3.8.7.2	X	X	X	4.2, 4.3.3 & 4.4
SPECIAL Switch	3.8.7.3	X	X	X	4.2, 4.3.3 & 4.4
OFFSET Switch	3.8.7.4	X	X	X	4.2, 4.3.3 & 4.4
BIT Switch	3.8.7.5	X	X	X	4.2, 4.3.3 & 4.4
DIS TYPE Switch	3.8.7.6	X	X	X	4.2, 4.3.3 & 4.4
AUDIO Control	3.8.7.7	X	X	X	4.2, 4.3.3 & 4.4
DMR Control	3.8.7.8	X	X	X	4.2. 4.3.3 & 4.4
Interface	3.8.8	X			4.2.3
Mounting	3.8.9	X			4.2.3
Form Factor	3.8.10	X		X	4.2.3 & 4.4
Contents	3.8.11	X			4.2.3
Electrical Connectors	3.8.12	X		X	4.2.3 & 4.4

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Table V. (U) Quality Conformance Inspection (Continued)

Requirement Description	Requirement Paragraph	First Article Tests	Production Acceptance Tests	Lot Sample Test	Test Paragraph
Cabling and Connections	3.8.13	X			4.2.3
Warm-up Time	3.7.3.4	X			4.2.5
Vibration	3.10.2.1	X		X	4.2.5 & 4.4
Temperature	3.10.2.2	X		X	4.2.5 & 4.4
Altitude	3.10.2.3	X		X	4.2.5 & 4.4
Shock	3.10.2.4	X			4.2.5
Humidity	3.10.2.5	X			4.2.5
Salt Fog Atmosphere	3.10.2.6	X			4.2.5
Explosive Atmosphere	3.10.2.7	X			4.2.5
Interference Control	3.10.2.8	X		X	4.2.4,
Total Weight	3.11	X		X	4.2.3 & 4.4
Nomenclature, Name-plates and Identi-fication Marking	3.13	X	X	X	4.2.3,4.3.3 & 4.4
Assembly, Subas-sembly and Component Identification	3.13.1	X	X		4.2.3 & 4.3.3
Workmanship	3.14	X	X	X	4.2.3 4.3.2 & 4.4

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shall be delivered by the Contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Para No.	Data Requirements Title	Applicable DID No.
3.2	Reliability program plan	DI-R-7079
3.4.1	Part approval request	DI-MISC-80071
3.5.1	Plan for maintaining configuration	DI-E-1100A
3.7	Safety assessment report	DI-SAFT-80102
3.7	System safety program plan	DI-SAFT-80100
3.7	System safety Hazard analysis report	DI-SAFT-80101
3.13	Request, Assignment of Nomenclature	UDI-E-21582
3.13	Request, Confirmation of Nomenclature	DI-CMAN-80194
3.13	Request, Approval of Identification Plate Drawings	DI-CMAN-80196
3.13	Request, Serial Number Assignment	DI-CMAN-80195
3.13	Request, Assignment of Type Equipment Code	UDI-E-21586
3.13	Request, Confirmation of Type Equipment Code	UDI-E-21587
4.1.5	Procedures, Test	UDI-T-21347
4.1.5	First article inspection procedure	DI-T-4901
4.1.5	First article inspection report	DI-T-4902
4.2	First article qualification test plan	DI-T-5315
4.2.1	Design data for equipment submitted for testing	UDI-E-21338
4.2.4	Plan, Control electromagnetic interference	DI-EMCS-80199
4.2.4	Plan, test electromagnetic interference	DI-EMCS-80201
4.2.4	Report, test electromagnetic interference	DI-EMCS-80200
4.2.5	Procedures reliability test	DI-RELI-80251
4.2.5	Reliability test plan	DI-RELI-80250
4.2.5	Reliability test report	DI-RELI-80252
4.2.5	report, failure	DI-R-21598
4.2.6	Report, reliability and maintainability program	UDI-R-21131
4.2.6	Report, reliability and maintainability test plan	UDI-R-21135
4.2.6	Report, reliability and maintainability test results	UDI-R-21136
4.3	Acceptance test procedures	DI-T-3714
4.3	Acceptance Test reports	DI-T-3716

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(Data item descriptions related to this specification and identified in Section 6 will be approved and listed as such in DOD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required the item should be a first article sample. The first article should consist of five units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and the bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence should apply:

a. Contract. The contract should have precedence over any specification.

b. This specification. This specification should have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced specifications. Any referenced specifications should have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications should apply to the extent specified.

6.5 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which should be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in MIL-E-5400, their use should be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

6.6 Type designations. The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.13. The correct type number should be used on nameplates, shipping records and instruction books, as applicable.

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6.7 Associated equipment. The equipment should operate with the following associated equipment comprising the AN/ALR-67(V)1 or AN/ALR-45F(V)2 Countermeasures Receiving Set as specified in 3.8.8.

a. CP-1293/ALR-67(V) Countermeasures Computer
b. IP-1276/ALR-67(V) Azimuth Indicator
c. R-2055/ALR-67(V) Radar Receiver (Special)
d. R-2148/ALR-67(V)1 Radar Receiver (Quadrant)
e. AS-3190/ALR-67(V) Antenna (Integrated Array)
f. AS-3189/ALR-67(V) Antenna (Quadrant)

6.8 Abbreviations and Acronyms. The following abbreviations and acronyms apply to this specification.

AC	Alternating Current
ACFT	Aircraft
ATE	Automatic Test Equipment
BIT	Built-in-Test
CDRL	Contract Data Requirements List
CI	Configuration Item
DC	Direct Current
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
ER	Established Reliability
FACI	First Article Configuration Inspection
GND	Ground
Hz	Hertz (cycles per second)
MTBF	Mean Time Between Failure
MTTR	Mean Time to Repair
OTP	Operational Test Procedure
PROM	Programmable Read Only Memory
PWR	Power
QRA	Quick Replaceable Assembly
RCVR	Receiver
RF	Radio Frequency
RFI	Radio Frequency Interference
RH	Relative Humidity
RMS	Root Mean Square
SRA	Shop Replaceable Assembly
V	Volt(s)
VA	Volt-Ampere(s)
VAC	Volts (Alternating Current)
VAST	Versatile Avionics Shop Tester
VDC	Volts (Direct Current)
WRA	Weapon Replaceable Assembly

6.9 Subject term (keyword) listing.

Countermeasures radar detection
Control equipment
Indicator lights, display devices

Preparing Activity
Navy-AS
(Project 5865-N062)

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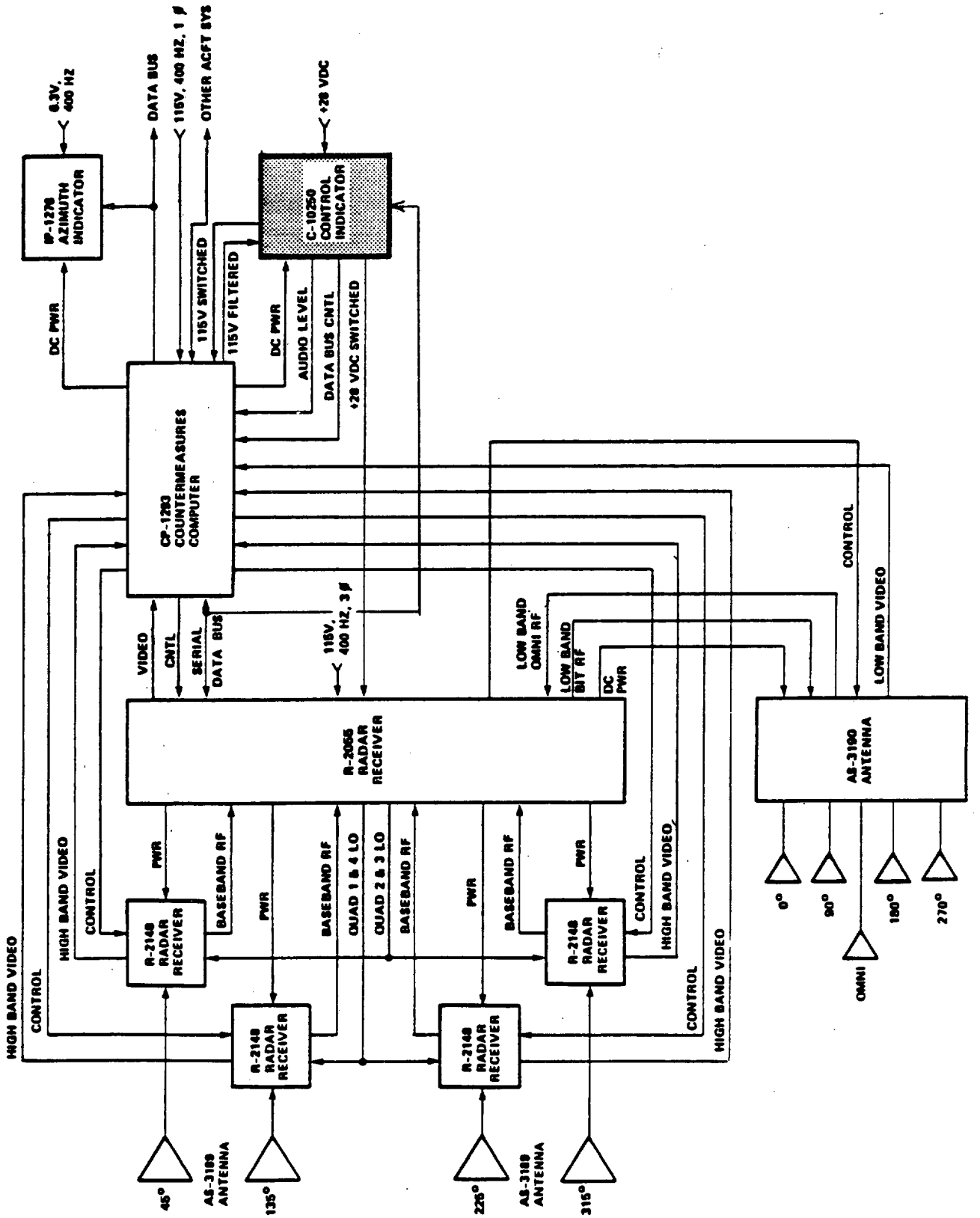


Figure 1 (U) AN/ALR-67(V)2 Countermeasures receiving set equipment configuration

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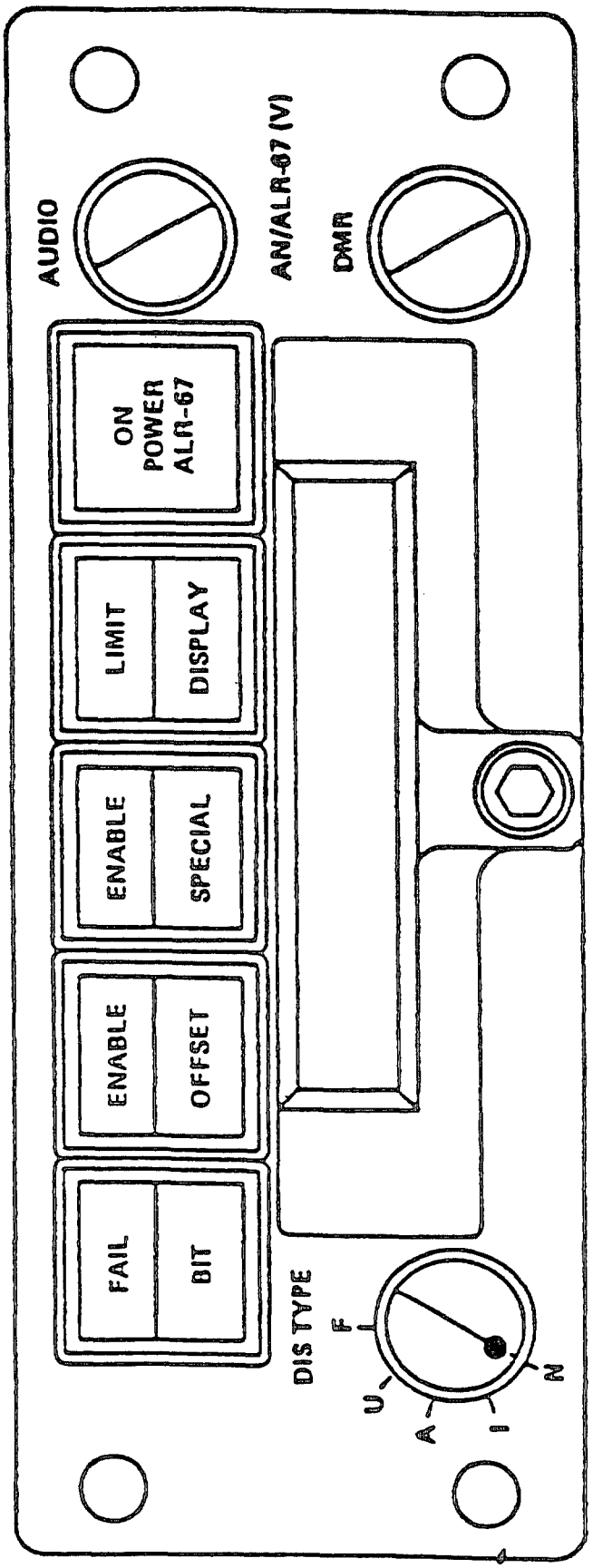


Figure 2 (U) Control panel layout diagram

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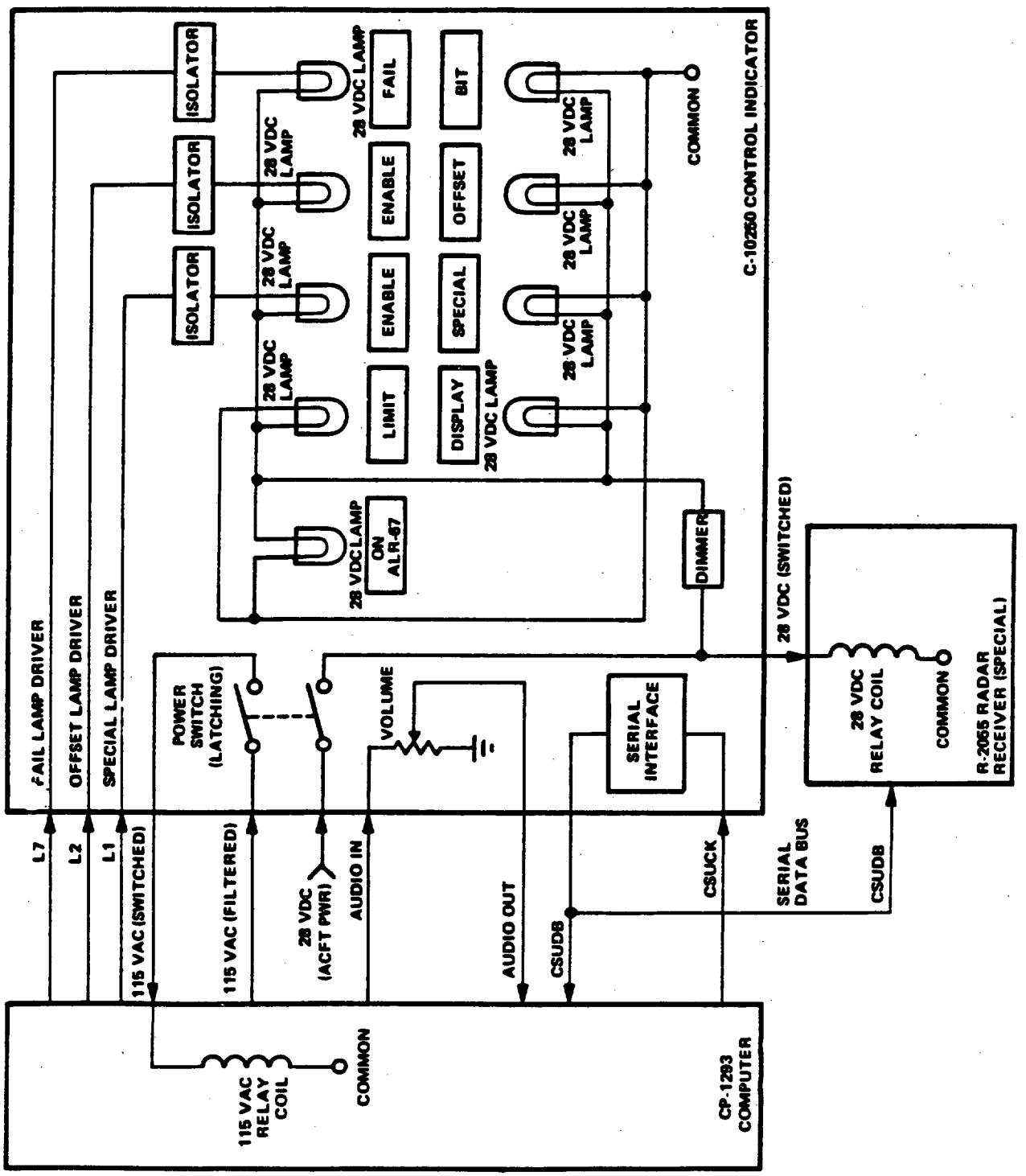


Figure 3 (U) Functional block diagram indicating, control

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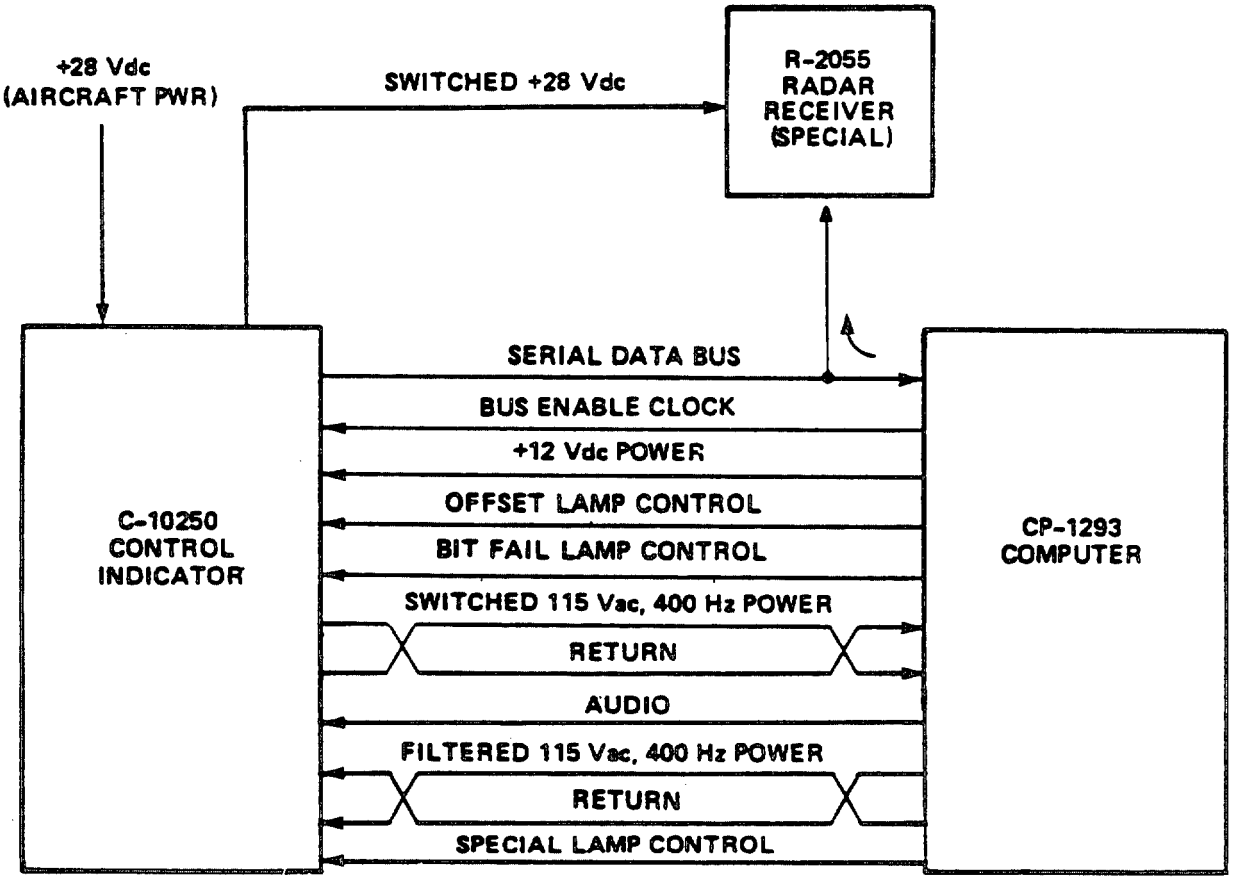


Figure 4 (U) Equipment/computer interface diagram

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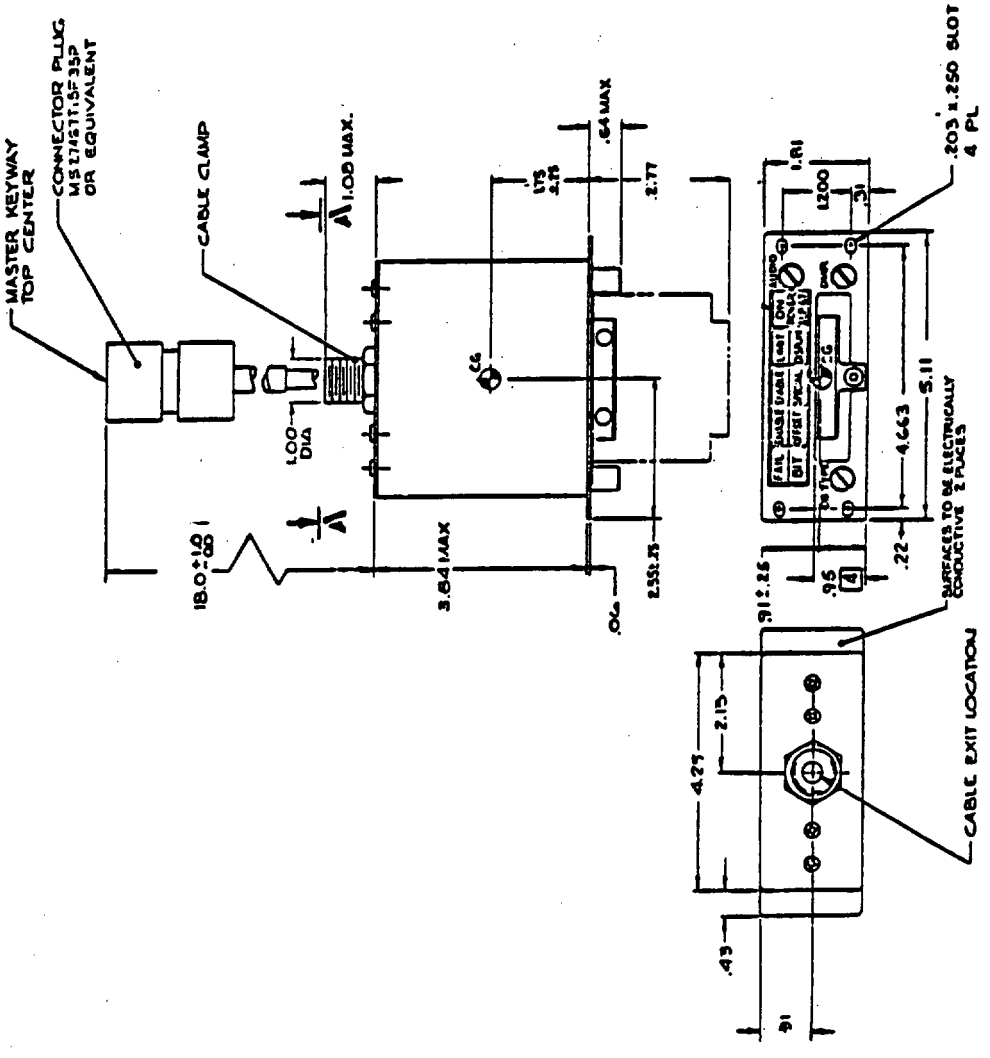


Figure 5 (U) Equipment form factor

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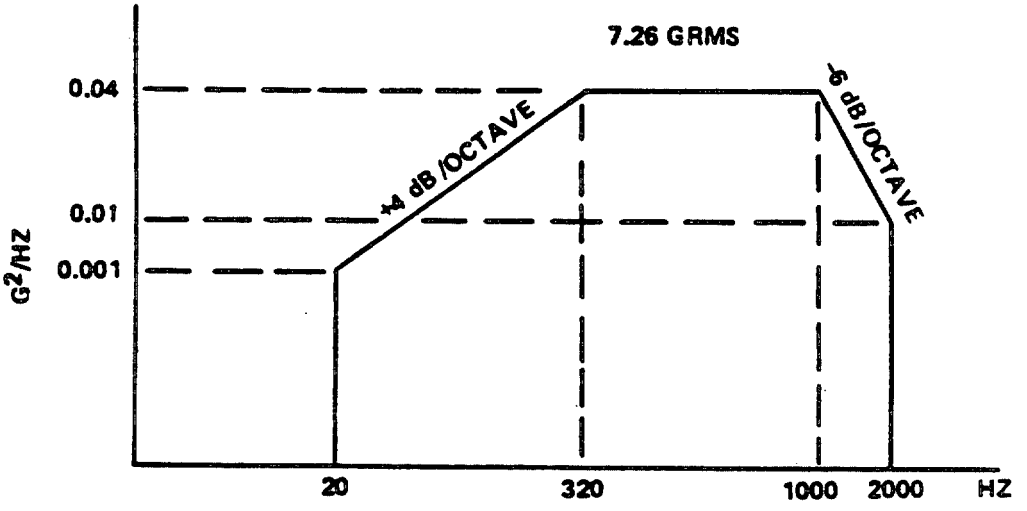
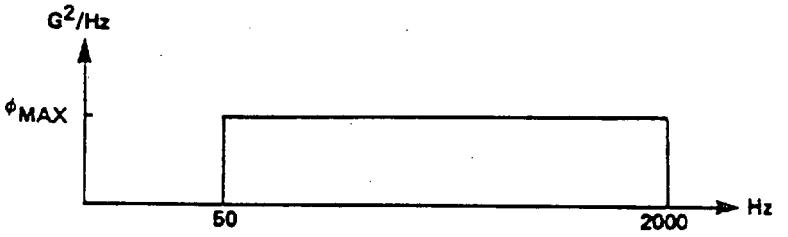


Figure 6 (U) Environmental qualification random vibration

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.9 GRMS FOR 56.3%	ϕ_{MAX} = .0004 G^2/Hz
2.0 GRMS FOR 34.2%	ϕ_{MAX} = .0020 G^2/Hz
4.4 GRMS FOR 9.5%	ϕ_{MAX} = .0099 G^2/Hz

ϕ_{MAX} = 0.0004 G^2/Hz RANDOM, 50 TO 2000 Hz, AT 0.9 GRMS FOR 56.3 PERCENT OF VIBRATION ON TIME.

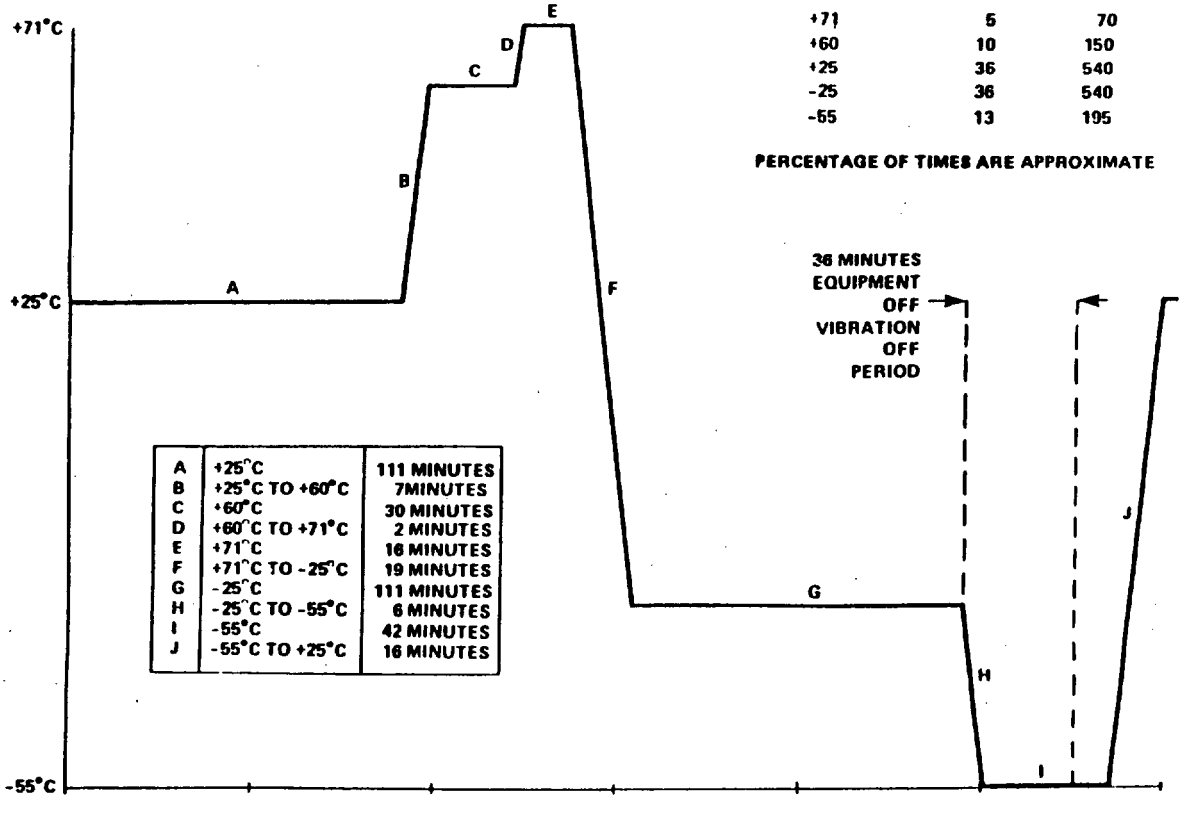
ϕ_{MAX} = 0.0020 G^2/Hz RANDOM, 50 TO 2000 Hz, AT 2.0 GRMS FOR 34.2 PERCENT OF VIBRATION ON TIME.

ϕ_{MAX} = 0.0099 G^2/Hz RANDOM, 50 TO 2000 Hz, AT 4.4 GRMS FOR 9.5 PERCENT OF VIBRATION ON TIME.

PERCENTAGE OF TOTAL TIME TOLERANCE $^{+0\%}_{-5\%}$

TEMPERATURE, °C	TIME, %	HOURS
+71	5	70
+60	10	150
+25	36	540
-25	36	540
-55	13	195

PERCENTAGE OF TIMES ARE APPROXIMATE



A	+25°C	111 MINUTES
B	+25°C TO +60°C	7MINUTES
C	+60°C	30 MINUTES
D	+60°C TO +71°C	2 MINUTES
E	+71°C	16 MINUTES
F	+71°C TO -25°C	19 MINUTES
G	-25°C	111 MINUTES
H	-25°C TO -55°C	6 MINUTES
I	-55°C	42 MINUTES
J	-55°C TO +25°C	16 MINUTES

Figure 7 (U) Reliability growth test profile

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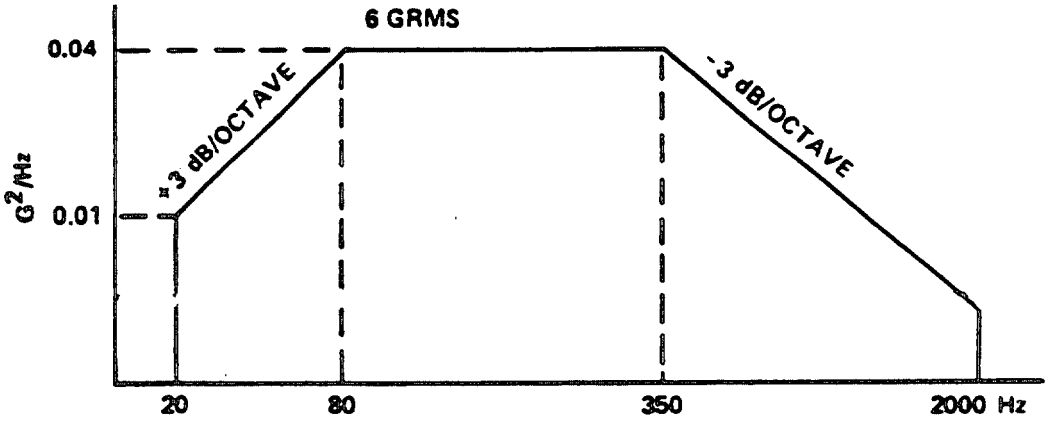


Figure 8 (U) Manufacturing screening random vibration

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-85569A(AS)	INDICATOR, CONTROL C-10250A/ALR-67(V)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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